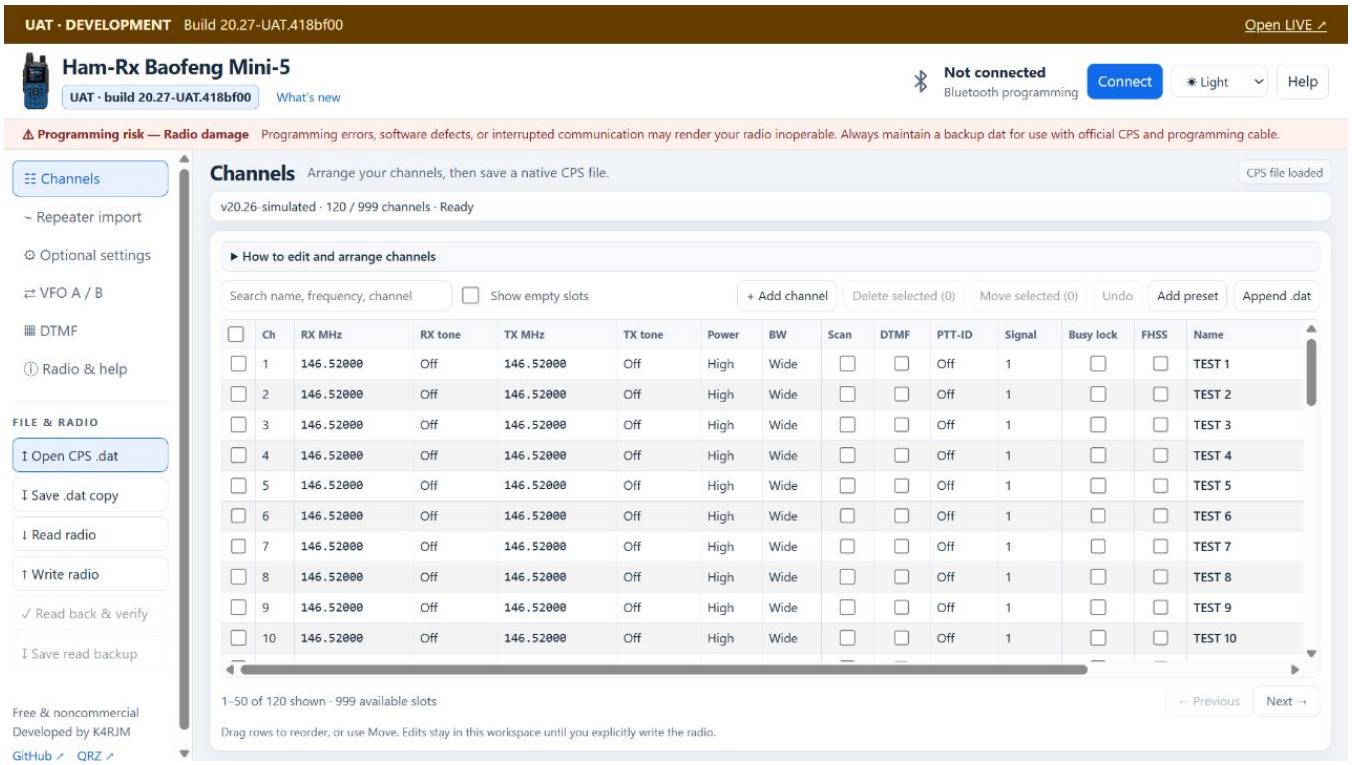
Start with a safe copy

Ham-Rx CPS User Guide **Version 20.27** | 19 October 2026
Free, noncommercial software developed by K4RJM.

Ham-Rx edits the MINI-5's channels and CPS settings in your browser. Editing, saving a file and programming a radio are three separate actions. This guide describes version 20.27. Screenshots were captured in UAT, the separately marked development site; LIVE uses the same controls.



Actual UAT 20.27 interface with demonstration channels. The data is synthetic; no radio was connected for these screenshots. Use PDF zoom to inspect small controls.

The normal workflow

1. Connect and read your radio, or open an existing native CPS .dat file.
2. Save an untouched backup. Then edit channels and settings.
3. Save the edited .dat under a different name.
4. Review the programming checklist, write only when ready, wait for the radio to restart, and read back to verify.

A native DAT can also be opened in stock Baofeng CPS with a compatible programming cable. Ham-Rx itself does not program over a cable or flash firmware. A DAT is a configuration backup, not a complete raw-memory or firmware image.

Connect, read and save

Prepare the radio and browser

Turn Wireless CPS programming ON on the radio. Stop scanning and close other programming applications. Keep the radio charged and within 3-6 feet (1-2 metres) of the computer. Use the HTTPS site in a browser that exposes Web Bluetooth. If Connect is unavailable, you can still edit native DAT files.

Read the current configuration

1. Select **Connect** and choose your MINI-5 in the browser Bluetooth chooser. The icon and text show the connection state. Connecting alone does not read or program the radio.
2. Select **Read radio**. Wait for the progress indicator to finish. A full read uses 521 blocks, even if few channels are populated.
3. Select **Save read backup** to download the original read as a native DAT. Confirm the file appears in Downloads and keep it unchanged.

Open and save a file

Open CPS .dat replaces the current workspace, including settings. If there are unsaved changes, a confirmation appears. Only native CPS DAT files are supported. For adding channels while keeping current settings, use Append .dat instead (page 5).

Save .dat copy downloads the current channels and CPS settings. Saving does not program the radio. Keep an original backup and a separate edited copy; check the downloaded file before closing or refreshing the page.

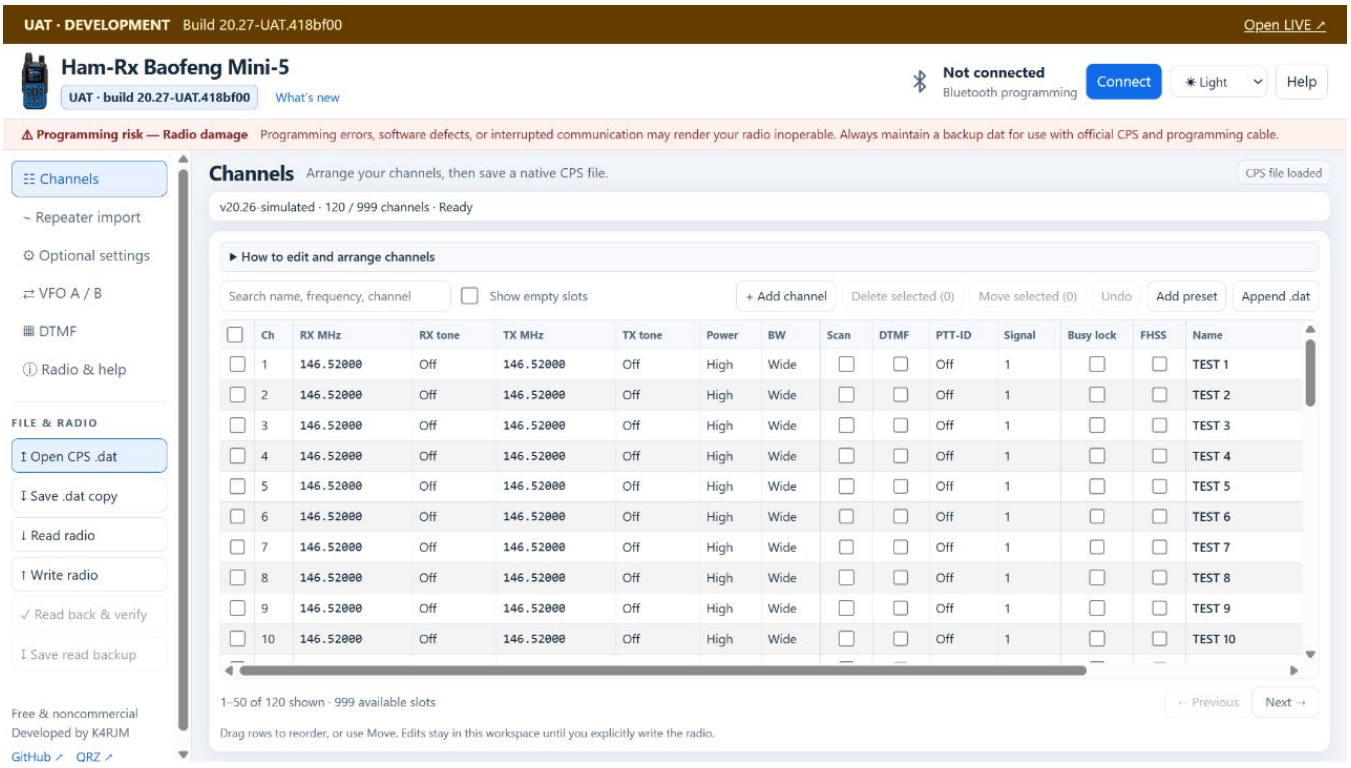
What stays on this device

The application holds files and radio captures in browser memory until downloaded. Only the appearance preference is stored locally. Refreshing or closing can lose unsaved work. The System, Light and Dark appearance choices are beside Connect.

Compatibility

The app checks the 5RMINI model identity for radio operations. A model match is not proof of firmware compatibility. Retained physical evidence includes Radio 100, firmware V0.05, hardware V1.7; it does not establish every MINI-5 variant. Read Testing & Compatibility in Radio & Help for the evidence and remaining limitations.

Edit and arrange channels



The channel table follows factory CPS field order. Scroll horizontally for fields beyond the visible width; the desktop table scrolls vertically within the workspace.

Edit in place: click a frequency, tone, power, bandwidth, PTT-ID, signal group or name cell. Use the checkboxes for Scan, DTMF, Busy lock and FHSS. Tab and Shift+Tab move between editors; Enter commits a text edit and Escape cancels the active edit. A dropdown selection applies to the workspace.

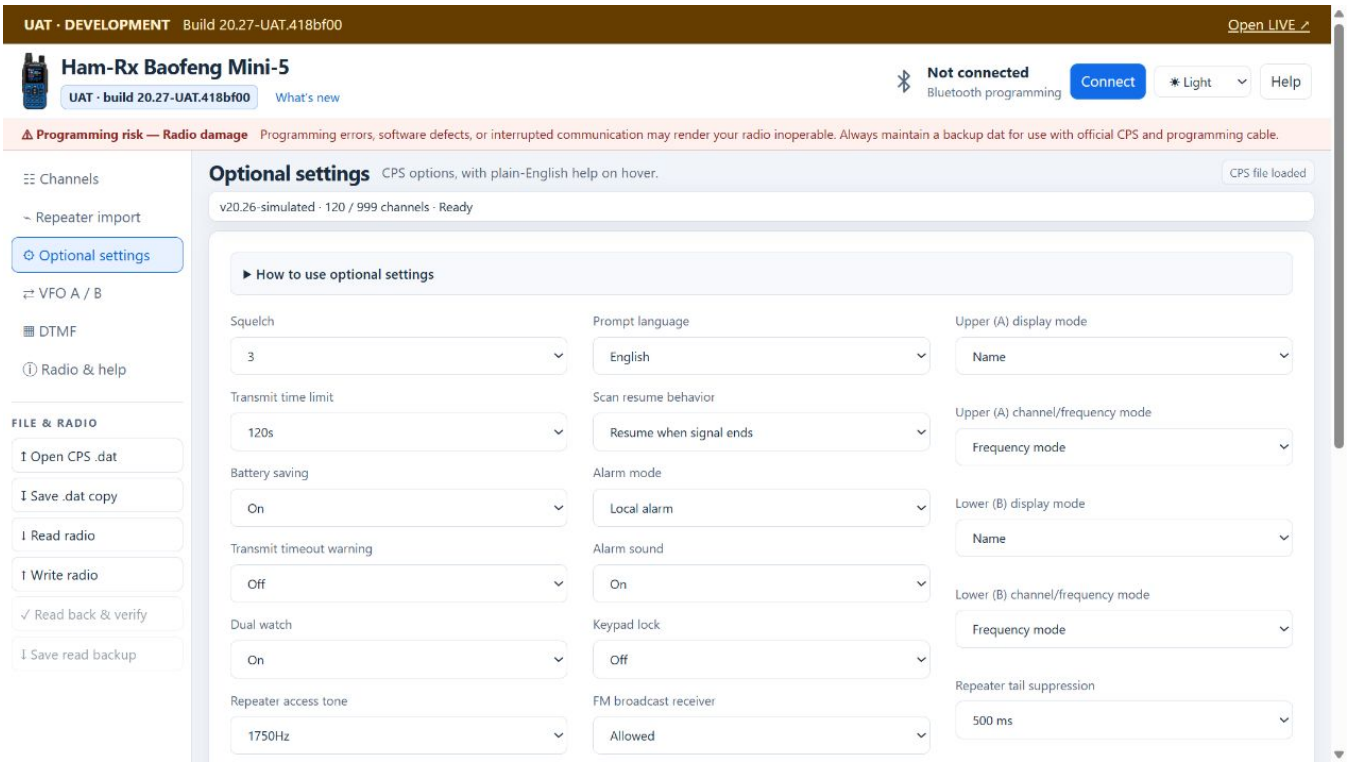
Frequencies such as **146.52** gain trailing zeroes. Memory-channel frequencies must fit the supported ranges and 1.25 kHz grid. The inline editor has two explicit boundary conversions: 174 becomes 173.99875, and 520 becomes 519.99875 MHz. Imported values are not automatically rounded.

Edit opens the full channel dialog. Previous / Next applies the current valid edits and moves between channels. Invalid entries stay in the current dialog for correction.

+ Add channel uses the first available empty slot. **Ins Ch** inserts above the chosen row and shifts later channels. **Move** shifts a channel to a destination. Select several row checkboxes and use **Move selected** for a contiguous move with a preview of displaced channel numbers.

Delete selected offers two choices: leave empty slots to preserve channel numbers, or remove gaps and renumber. **Undo** reverses the previous channel operation. Search and Show empty slots help locate channels; Previous / Next beneath the table changes the displayed page.

Optional settings, VFO and DTMF



Optional settings are edited in the workspace. Changing a field does not immediately change the radio.

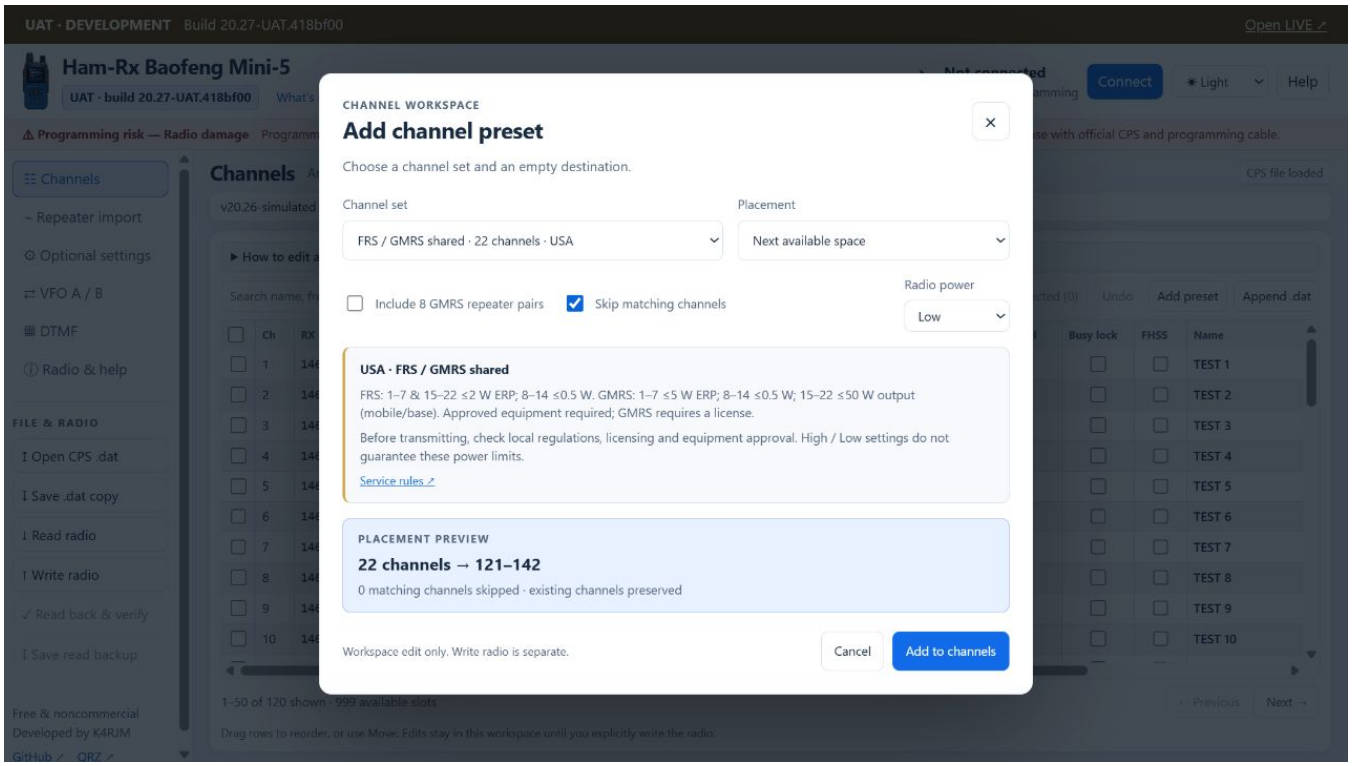
Optional settings includes squelch, scan resume behavior, display choices, PTT ID delay, VOX and Side-key short press. SK1 is the programmable side key: FM broadcast radio, Channel scan, Frequency search, VOX, Flashlight or Alarm. Keep unfamiliar settings at their existing values until you understand their effect.

VFO A / B controls the upper and lower frequency-mode settings. These are separate from stored memory channels. Review the limitation notice: offset direction and DTMF group share a representation in some combinations. On tested VFO A, OFF/group 17, PLUS/group 1 and PLUS/group 17 can all read back as PLUS/group 17. Use explicit RX and TX memory-channel frequencies for repeaters when appropriate.

DTMF edits signaling codes and groups. Channel PTT-ID and signal-group selections determine how configured codes are associated with a channel. These fields are not needed for ordinary voice channels unless the intended system requires them.

The factory CPS has separately displayed defaults for some stored PTT-ID/VFO values. Independent Bluetooth captures retained the intended values; a later factory write reset a delay. This observation must not automatically be blamed on the Bluetooth CPS or radio firmware.

Append channels without overwriting



The placement preview shows the destination and matching channels skipped before anything is added. This example starts after 120 demonstration channels.

Add preset offers the existing FRS/GMRS shared channels, GMRS repeater pairs, MURS and PMR446 sets. Read the service notice in the dialog. A frequency preset and a High/Low setting do not establish permission to transmit or equipment compliance.

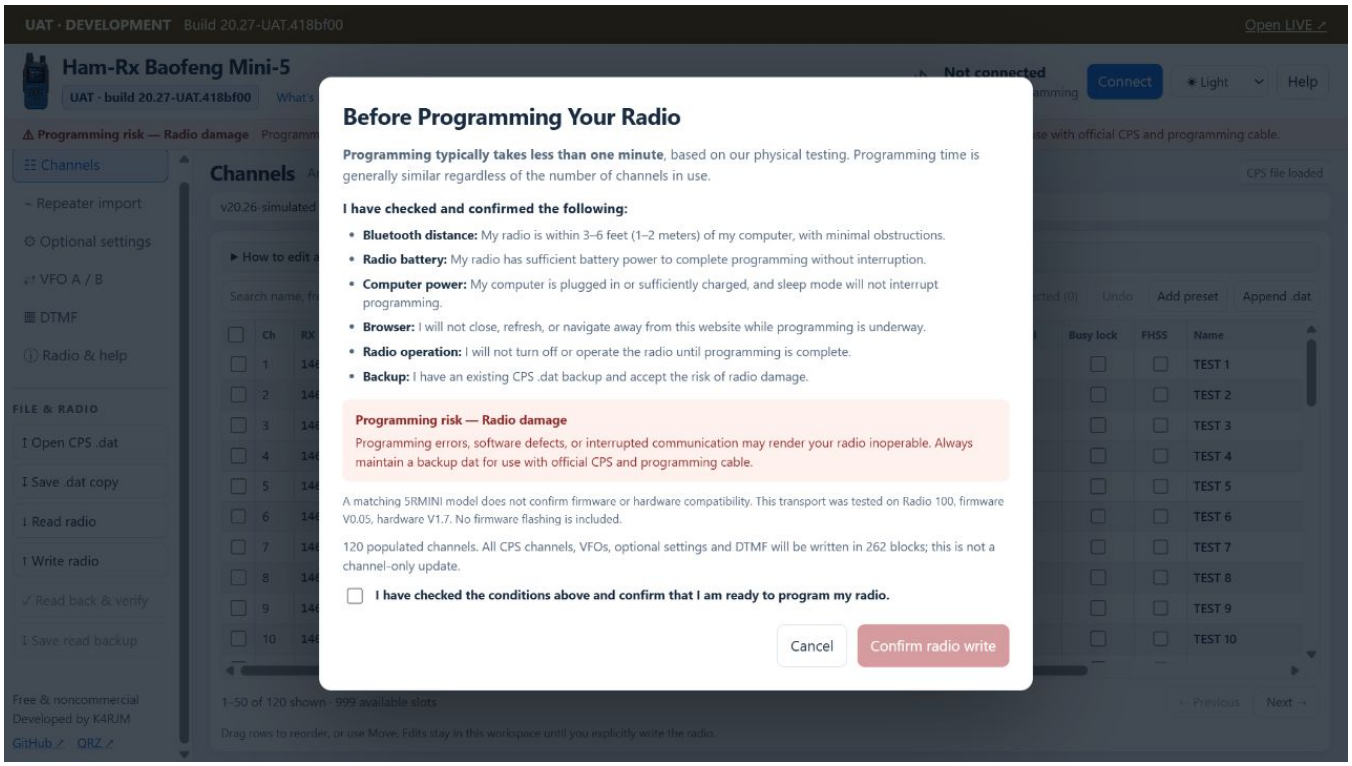
Append .dat takes populated channels from another native DAT and adds them to the current workspace. It retains the current global and VFO settings. This is useful after a radio read when you want to add a saved channel set rather than replace the whole configuration.

Choose **Next available space** or an available numbered block, starting at 101, 201 and so on through 901. Occupied blocks are identified. Review the proposed range, matching-channel count and placement errors. Existing populated channels are preserved; the app requires enough empty space.

Use **Cancel** to leave the workspace untouched or **Add to channels** to accept the preview. Then review the new rows, save a DAT copy and program separately if desired. Appending does not perform a partial radio write.

Repeater import is a preview pending RepeaterBook API approval. It does not fetch repeater data or accept shared tokens.

Program and verify



The actual confirmation window. This screenshot was captured and cancelled without connecting to or writing a radio.

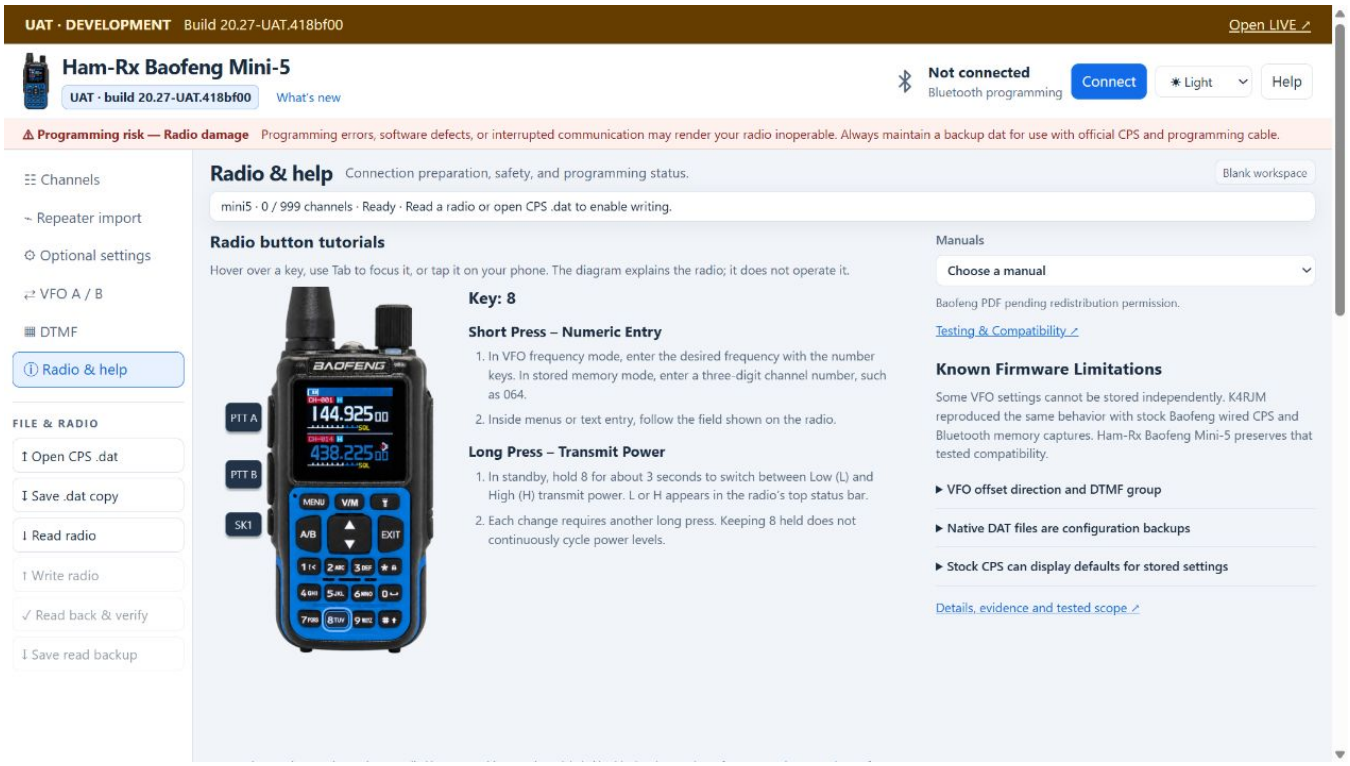
Select **Write radio** only after reviewing the entire workspace and keeping a backup. The single checkbox confirms the preparation conditions, including that you have a backup; it does not check your downloaded files. Cancel returns without programming.

Keep the radio nearby and charged, keep the computer awake, and do not operate the radio, navigate away or close the browser. Programming typically took less than a minute in physical tests. The app writes the full CPS configuration in 262 blocks, including channels, VFOs, optional settings and DTMF; channel count does not make it a partial update.

After the blocks are acknowledged, the app requests a browser disconnect and the radio normally reboots. Wait until it finishes. Select **Read back & verify**, then **Radio has restarted - reconnect**. If necessary, toggle Wireless CPS OFF then ON and use the fallback button. There is no fixed reboot timer.

Acknowledgements mean the transfer finished; readback is a separate verification. Check the intended settings on the radio as well. If a write times out or stops, its result is uncertain. The app does not automatically retry it. Save diagnostics and investigate before attempting another write.

Radio tutorials and reference documents



Radio & Help: interactive radio tutorials on the left, expandable limitations and the Manuals selector on the right.

Hover over a radio button, focus it with the keyboard, or tap it on a phone. Its short-press and documented long-press instructions appear beside the image. All 23 hotspots remain available.

Known Firmware Limitations opens each explanation separately. It retains the distinction between the shared VFO representation, factory-compatible DAT reconstruction and the stock-CPS display/write observation. The heading does not establish that every item is a firmware defect.

Choose **Ham-Rx CPS User Guide** in Manuals to read this PDF on the right. **Expand left** uses the larger column; **Return right** restores the smaller view. **Close manual** restores tutorials and limitations. The same PDF frame is retained to preserve reading position where browser support permits. **Open PDF in new tab** is also available.

The Baofeng MINI-5 Radio Manual is listed as **pending permission**. The manufacturer PDF is not hosted until redistribution permission is confirmed. Testing & Compatibility opens a separate site-hosted document with physical testing, automated checks, limitations and audit references.

Troubleshooting and evidence

Connect is unavailable or the radio is missing

Check that the page uses HTTPS and the browser offers Web Bluetooth. Turn Wireless CPS ON, stop scanning, keep the radio close and close competing programming apps. A browser without Bluetooth support can still edit and save DAT files. Availability depends on the browser and operating system.

Reconnect fails after programming

Wait for the radio to complete its reboot, then use the explicit reconnect prompt. Try its Wireless CPS OFF/ON fallback if needed. Some tests required toggling Bluetooth on both the PC and radio. That intermittent symptom is not established as fixed. Save your DAT and the diagnostic report before reloading if a browser call remains stuck.

An edit, save or write is blocked

Read the displayed validation message. Check channel frequency ranges and spacing, names and required fields. Opening a DAT does not verify radio identity. Writing stays locked after a write attempt for that source, and an uncertain write is never retried automatically. Do not bypass these protections.

Readback differs from what you expected

Compare the intended change, the readback and the Known Firmware Limitations. Do not treat accepted VFO collisions as proof of arbitrary memory corruption. A matching capture does not prove every operating function works; verify the intended operation on the radio.

Save useful evidence

In Radio & Help, choose **Save diagnostic report**. It contains connection logs and available captures for analysis. Review its contents before sharing. Preserve the original DAT, edited DAT, diagnostic report, version number and a concise description of what happened.

Testing and source

Physical testing by K4RJM supports the retained radio behavior. Version 20.27 is an interface/documentation update; no physical-radio write was performed while preparing it. Automated tests and simulated browser workflows supplement the physical evidence; they do not replace it.

Testing & Compatibility | Audit dispositions | UAT source

Developed by K4RJM. Derived from Mini 5 Studio by replicationbench, with protocol comparisons against CHIRP and native CPS captures. Software: GPL v3. This guide documents the Ham-Rx application and is not the manufacturer's radio manual.